

STAT 231 Midterm 1 Information

Please read the first page of this document in full before attempting either the practice or real Midterm.

Examination Logistics

Seating for Midterm 1 is predetermined, please check your seat assignment at <https://odyssey.uwaterloo.ca/teaching/schedule> near the date of the exam. **Important:** We are a large class and will be spread across several rooms, make sure you know which room your seat is in - it will not be possible for you to write the exam in a room other than the one to which you have been assigned!

Important: The exam start time on Odyssey is listed as 4:30pm but the exam may start slightly later if more time is required for setup. You should arrive before 4:30pm and wait outside the lecture theatre until you are called in.

The exam will last 90 minutes and divided into three parts: A, B, and C, as described in the Course Outline. You can find a question/marks breakdown on the last page of this document!

What to bring:

- Your Watcard. This will be checked during the exam.
- A non-programmable, faculty-approved calculator.
- A pen and/or pencil, and some kind of ruler/straight-edge.
- Any notes you wish to prepare (see below).

Calculations

Note that a calculator is required. However, unless a question says a numeric calculation or value is required, numeric answers left as numeric expressions will still receive full credit. (For example: an answer of $'1 + 1'$ would receive the same credit as an answer of $'2'$, unless the question says a numeric calculation or value is required.)

Probabilistic Expressions

A question may ask you to leave your answer as a 'probabilistic expression'. If this instruction is given - and only if this instruction is given - you may leave your answer in terms of a random variable with a clearly specified probability distribution. For example, if you were told that $X \sim G(1, 2)$ and $Y \sim G(3, 2)$ independently, and asked "Find $P(X + Y > 3)$. You may leave your answer as a probabilistic expression.", then a valid answer would be "Let $W = X + Y$, then $P(X + Y > 3) = P(W > 3)$ where $W \sim G(4, \sqrt{8})$."

Note that a 'probabilistic expression' and a 'numeric expression', as defined for STAT 231 exams, are distinct. For example, if you were asked "Let $Y \sim \text{Poisson}(3)$, find a numeric expression for $P(Y < 3)$.", it would *not* be acceptable to write "The answer is $P(Y < 3)$ where $Y \sim \text{Poisson}(3)$ ". Rather, your answer should be a numeric expression such as
$$\sum_{y=0}^2 \frac{3^y e^{-3}}{y!}.$$

Coverage

Midterm 1 covers the material in Chapters 1 and 2 of the Course Notes, except for Section 2.4 and Q-Q plots. It also covers material covered in Tutorials 1 and 2, the Introductory R Tutorial, R Tutorial 1, and Assignment 1.

Preparing for the Midterm

Along with the materials detailed under 'Coverage', you should review end-of-chapter problems from the Course Notes (as detailed on LEARN). There is a Practice Midterm available on LEARN, along with solutions.

Bringing Notes to the Exam

Our exams are not tests of memorization. You may bring any notes you wish to the exam. However, you are *not* permitted to bring or use any electronic devices (other than a calculator and/or a device whose primary function is telling time). As such, you may wish to prepare a set of reference notes to bring with you. These may be in any format you wish (such as handwritten on loose-leaf paper, in a notebook, or typeset and printed).

Your notes may be as detailed as you wish, but we would advise keeping your notes as concise as possible (it is not particularly efficient to be rifling through piles of notes during an exam).

To aid you in preparing for the exam, here is a list of the ideas and definitions you should know:

Empirical Studies (Sections 1.1-1.2)

- Units, populations and processes.
- Variate and types of variates. Response versus explanatory variates.
- Attributes of a population or process.
- Types of studies (sample surveys, observational studies, experimental studies).

Numerical Summaries (Section 1.3)

- Measures of location: sample mean, median, mode.
- Measures of variability: sample variance, sample standard deviation, range, IQR.
- Measures of shape: skewness, kurtosis.
- Sample percentiles and quantiles. Lower or first quartile, upper or third quartile. Five number summary.
- Numerical summaries for bivariate data: sample correlation, relative risk.

Graphical Summaries (Section 1.3)

- Relative frequency histograms.
- Empirical cumulative distribution function.
- Boxplots, scatterplots, run charts.

Data Analysis and Statistical Models (Sections 1.4-1.5, 2.1)

- Descriptive statistics.
- Discrete statistical models: $\text{Binomial}(n; \theta)$, $\text{NegativeBinomial}(k; \theta)$, $\text{Poisson}(\theta)$, $\text{Geometric}(\theta)$.
- Continuous statistical models: $\text{Exponential}(\theta)$, $G(\mu, \sigma)$.

Estimation (Sections 2.2, 2.3, 2.5)

- Estimate of a parameter.
- Likelihood functions for discrete and continuous distributions.
- Maximum likelihood estimate (Definition 12).
- Relative likelihood function (Definition 13) and log likelihood function (Definition 14).
- Likelihood function for a random sample.
- Invariance property of maximum likelihood estimates (Theorem 16).

Check the Fit of a Model (Section 2.6)

You should know how to check the fit of a model by:

- Calculating expected frequencies using the assumed model (discrete distributions **only**) and comparing them with observed relative frequencies.
- Examining a graph of the probability density function of the assumed model superimposed on a relative frequency histogram for the observed data (continuous distributions).
- Examining a graph of the cumulative distribution function of the assumed model superimposed on a graph of the empirical cumulative distribution function for the observed data (continuous distributions).

R Commands

You should be familiar with all R commands introduced in the Introductory R Tutorial and R Tutorial 1, and any functions used in lectures and tutorials. You may be given R commands and the resulting output (numerical and/or graphical) and be asked to interpret it. You may be asked what the output would be from running a particular R command on a sample of data. You may be asked which of a list of R commands may be appropriate for a particular analysis. You may be asked to identify errors in R commands.

Exam Front Page Instructions

For your reference, the following information will be provided on the front page of your examination booklet. Note that as this is an open book exam, no formula pages will be provided.

- This exam consists of **three parts**. The mark breakdown is as follows:

Part	A	B	C			
Question	1	2	3	4	5	Total
Marks	15	15	10	10	10	60

- **Unless a question indicates a numeric value is required**, numeric answers may be left as numeric expressions if desired (e.g., if the correct answer was 2, an answer of ‘1 + 1’ would receive the same credit as an answer of ‘2’). If you express answers as numeric values these should be accurate to at least three decimal places.
- If you are asked to identify the type of a variate, choose from Continuous, Discrete, Categorical, Ordinal, or Complex.
- If you believe insufficient information is provided in a question to determine the requested answer, you may write ‘not enough information’, ‘NEI’, or similar. If the question indicates you should justify your answer, this justification is still required even if your answer is ‘not enough information’.
- Additional blank pages are provided at the end of the exam for rough work and/or additional space for answers. If you wish any work on these pages to be graded, you must clearly indicate this on the corresponding question page.
- Good luck :)